

PNP Epitaxial Silicon Transistor

SS8550

Features

- 2 W Output Amplifier of Portable Radios in Class B Push-Pull Operation
- Complementary to SS8050
- Collector Current: $I_C = 1.5$ A
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Parameter	Symbol	Value	Unit
Collector-Base Voltage	V_{CBO}	-40	V
Collector-Emitter Voltage	V_{CEO}	-25	V
Emitter-Base Voltage	V_{EBO}	-6	V
Collector Current	I_C	-1.5	A
Junction Temperature	T_J	150	$^\circ\text{C}$
Storage Temperature	T_{STG}	-65 to 150	$^\circ\text{C}$

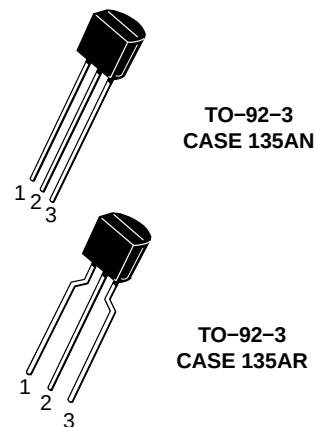
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS (Note 1)

($T_A = 25^\circ\text{C}$ unless otherwise noted)

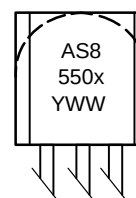
Parameter	Symbol	Value	Unit
Power Dissipation	P_D	1	W
Power Dissipation Derate Above 25°C	P_D	8	mW/ $^\circ\text{C}$
Thermal Resistance, Junction-to-Ambient	$R_{\theta JA}$	125	$^\circ\text{C/W}$

1. PCB size: FR-4, 76 mm x 114 mm x 1.57 mm (3.0 inch x 4.5 inch x 0.062 inch) with minimum land pattern size.



1. Emitter
2. Base
3. Collector

MARKING DIAGRAM



S8550x = Specific Device Code
 Line 1: A = Assembly Location
 Line 2: x = C or D
 Line 3: Y = Year
 WW = Work Week

ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

SS8550

ELECTRICAL CHARACTERISTICS ($T_C = 25^\circ\text{C}$ unless otherwise noted)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
BV_{CBO}	Collector–Base Breakdown Voltage	$I_C = -100\ \mu\text{A}$, $I_E = 0$	-40			V
BV_{CEO}	Collector–Emitter Breakdown Voltage	$I_C = -2\ \text{mA}$, $I_B = 0$	-25			V
BV_{EBO}	Emitter–Base Breakdown Voltage	$I_E = -100\ \mu\text{A}$, $I_C = 0$	-6			V
I_{CBO}	Collector Cut–Off Current	$V_{CB} = -35\ \text{V}$, $I_E = 0$			-100	nA
I_{EBO}	Emitter Cut–Off Current	$V_{EB} = -6\ \text{V}$, $I_C = 0$			-100	nA
h_{FE1}	DC Current Gain	$V_{CE} = -1\ \text{V}$, $I_C = -5\ \text{mA}$	45	170		
h_{FE2}		$V_{CE} = -1\ \text{V}$, $I_C = -100\ \text{mA}$	85	160	300	
h_{FE3}		$V_{CE} = -1\ \text{V}$, $I_C = -800\ \text{mA}$	40	80		
$V_{CE(sat)}$	Collector–Emitter Saturation Voltage	$I_C = -800\ \text{mA}$, $I_B = -80\ \text{mA}$		-0.28	-0.50	V
$V_{BE(sat)}$	Base–Emitter Saturation Voltage	$I_C = -800\ \text{mA}$, $I_B = -80\ \text{mA}$		-0.98	-1.20	V
$V_{BE(on)}$	Base–Emitter On Voltage	$V_{CE} = -1\ \text{V}$, $I_C = -10\ \text{mA}$		-0.66	-1.00	V
C_{ob}	Output Capacitance	$V_{CB} = -10\ \text{V}$, $I_E = 0$, $f = 1\ \text{MHz}$		15		pF
f_T	Current Gain Bandwidth Product	$V_{CE} = -10\ \text{V}$, $I_C = -50\ \text{mA}$	100	200		MHz

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

h_{FE} CLASSIFICATION

Classification	C	D
h_{FE2}	120 ~ 200	160 ~ 300

ORDERING INFORMATION

Part Number	Top Mark	Package	Shipping
SS8550CBU	S8550C	TO-92-3, case 135AN (Pb-Free)	10,000 Units/ Bulk Box
SS8550CTA	S8550C	TO-92-3, case 135AR (Pb-Free)	2,000 Units/ Fan-Fold
SS8550DBU	S8550D	TO-92-3, case 135AN (Pb-Free)	10,000 Units/ Bulk Box
SS8550DTA	S8550D	TO-92-3, case 135AR (Pb-Free)	2,000 Units/ Fan-Fold

TYPICAL PERFORMANCE CHARACTERISTICS

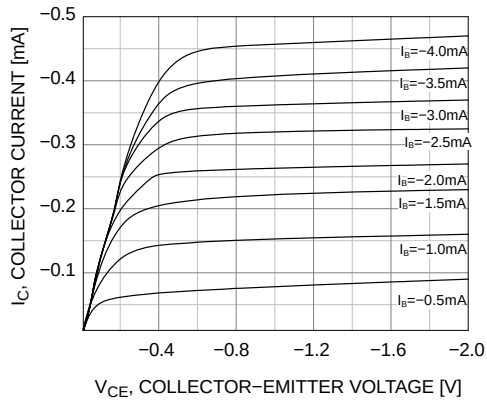


Figure 1. Static Characteristic

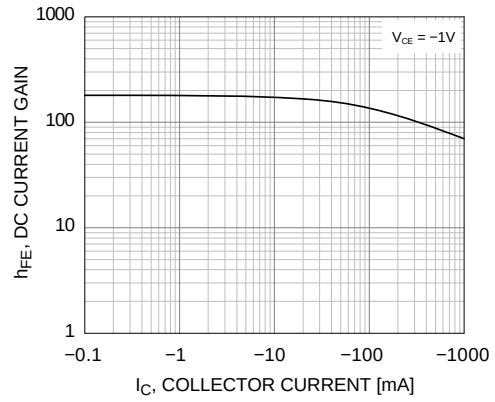


Figure 2. DC Current Gain

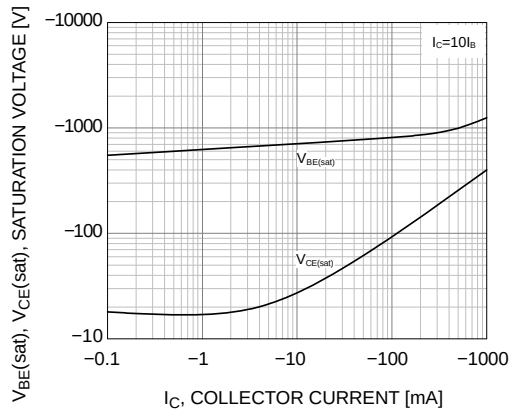


Figure 3. Base-Emitter Saturation Voltage and Collector-Emitter Saturation Voltage

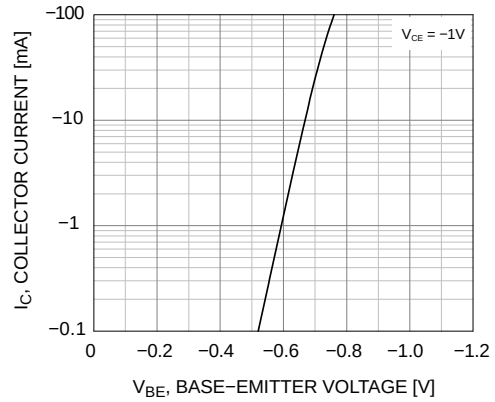


Figure 4. Base-Emitter On Voltage

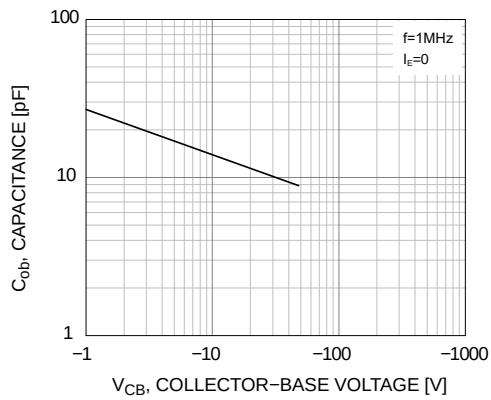


Figure 5. Collector Output Capacitance

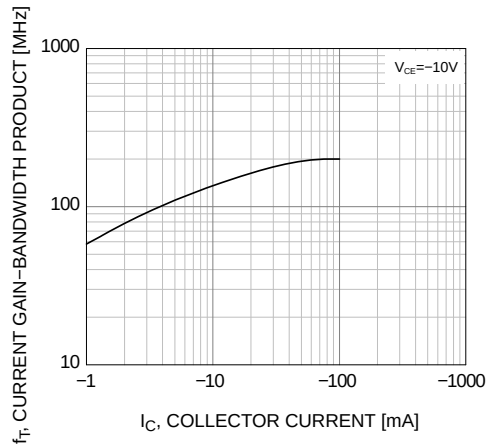
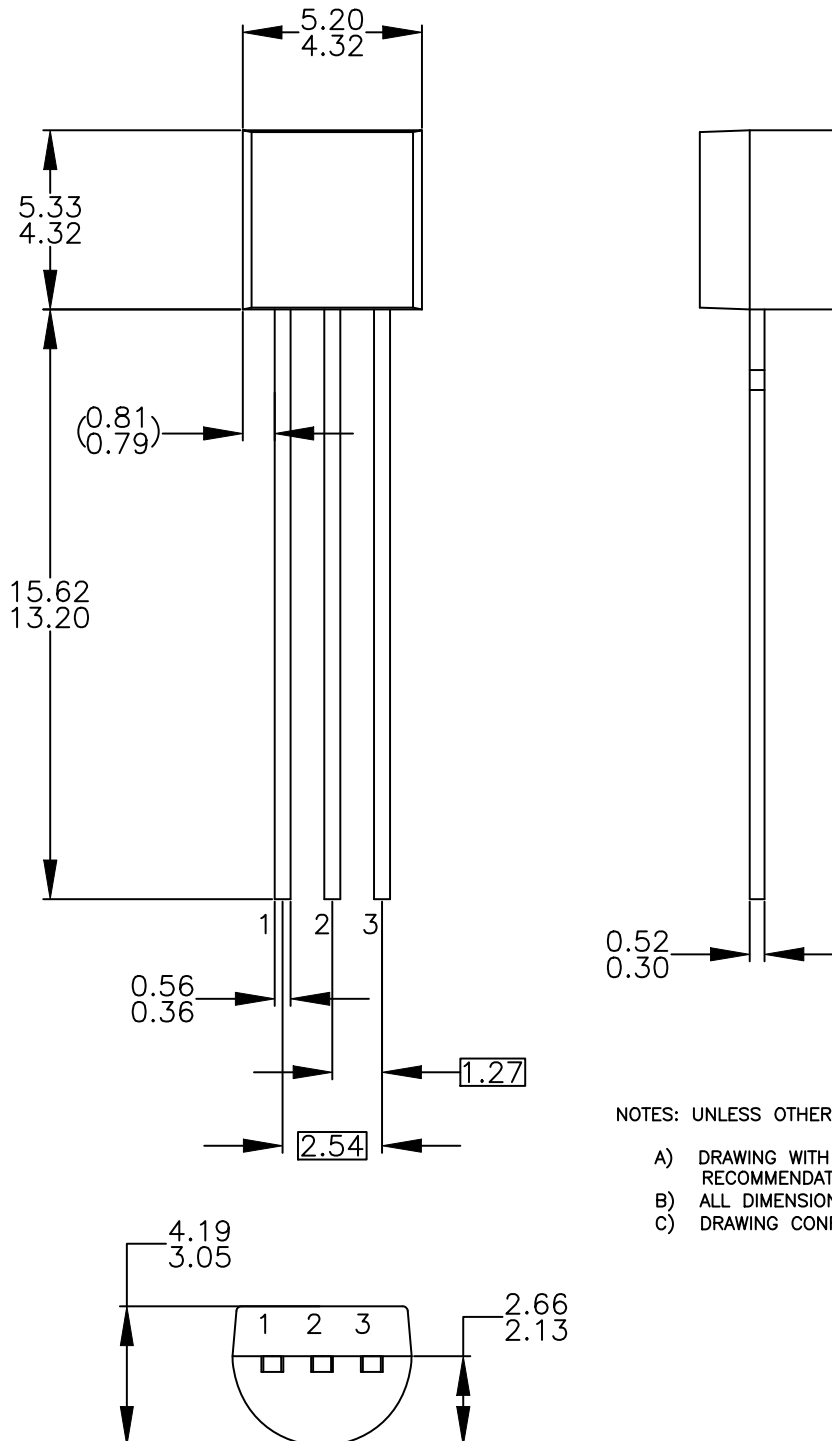


Figure 6. Current Gain Bandwidth Product

TO-92 3 4.825x4.76
CASE 135AN
ISSUE O

DATE 31 JUL 2016



NOTES: UNLESS OTHERWISE SPECIFIED

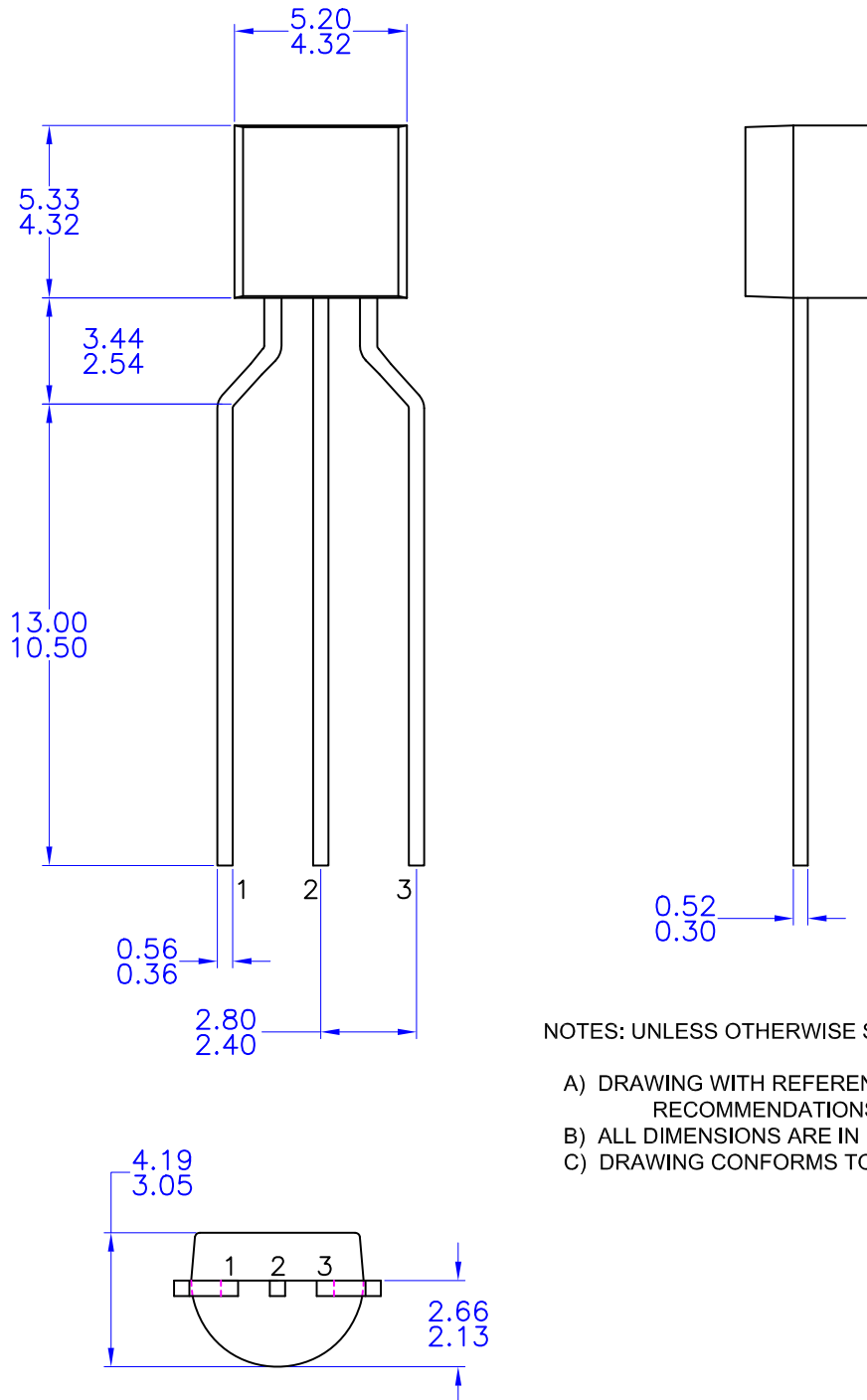
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